

Program : Diploma in Tool & Die Engineering	
Course Code : 3118	Course Title: Conventional Machine Tools Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide knowledge about the working of Drilling machine, shaper, slotter, milling and grinding machines.
- To familiarize various measuring instruments for taking dimension.
- To familiarize students with different operations on lathe, drilling, shaper, slotter, milling and grinding machines.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic Bench work operations		Production Engineering I	2
Identification of lathe parts		Production Engineering II	3
Basic filing operations		Engineering Workshop practice	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Perform operations on shaping machine.	12	Applying
CO2	Demonstrate various operations on milling machines/Slotter	12	Applying
CO3	Demonstrate the operations of different grinding machines.	9	Applying
CO4	Practice precise boring operations on tool room lathe.	9	Applying
	Lab Exam	3	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3			3			3
CO3	3			3			
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Perform operations on shaping machine.		
M1.01	Perform shaping operations like flat, step, channel and angular surfaces.	12	Applying
CO2	Demonstrate various operations on milling machines/Slotter.		
M2.01	Perform plain, step and channel milling operation using different milling cutters.	6	Applying
M2.02	Illustrate Keyway cutting operation on slotter.	6	Applying
	Lab Exam – I	1.5	
CO3	Demonstrate the operations of different grinding machines		
M3.01	Practice on surface grinding/cylindrical grinding machines.	6	Applying
M3.02	Demonstrate grinding wheel grades and selection Mounting, balancing and dressing of grinding wheel.	3	Applying
CO4	Practice precise boring operations on tool room lathe		
M4.01	Demonstrate Four jaw chuck setting using dial indicators, Boring holes to an accuracy of +/- 0.02 mm, reaming with machine reamer, check with plug gauge.	6	Applying
M4.02	Drill deep through holes.	3	Applying
	Lab Exam - II	1.5	

Text/ Reference:

T/R	Book Title/Author
T1	Mechanical Workshop Practice by K. C John, PHI Learning Private Limited
R2	Mechanical Workshop & Laboratory Manual by K. C. John
R3	Workshop Technology Vol. I by S K HajraChoudhary
R4	Workshop Technology Vol. II by S K HajraChoudhary

Online Resources:

Sl.No	Website Link
1	http://www.vssut.ac.in/lecture_notes/lecture1423905443.pdf
2	https://www.dphu.org/uploads/attachements/books/books_3017_0.pdf
3	https://nptel.ac.in/content/storage2/courses/112105127/pdf/LM-20.pdf
4	https://science.jrank.org/pages/4045/Machine-Tools.html